

Upper irredundance equals independence in direct products of complete multipartite graphs: a proof of Conjecture 1.2 of Alon and Defant

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Abstract

Let $H_1, \dots, H_n$ be finite complete multipartite graphs and let $G = \prod_{i=1}^n H_i$ be their direct (tensor) product. We prove that the upper irredundance number of G equals its independence number, $\text{IR}(G) = \alpha(G)$, with no balance assumption on the factors. This is Conjecture 1.2 of Alon and Defant, who proved the balanced case conjectured by Burcroff in all but 37 explicitly listed products; the present argument covers those 37 products as well, and also gives $\alpha(G) = \Gamma(G)$ for the upper domination number, as conjectured by Defant and Iyer for balanced factors. The proof refines the Alon–Defant bound $|\text{Soc}(S)| \leq 2^n$ on the number of social vertices of an irredundant set S to $|\text{Soc}(S)| + 2\rho \leq 2^n$, where ρ is the rank of a tensor representation of the lonely vertices, and matches it with a weight-deficit inequality $\alpha(G) - W(\mathcal{A}) \geq 2^n - 2\rho$ for independent label families $\mathcal{A}$, proved by a rank-nonincreasing compression to a down-set and a count of maximal intersecting families.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 5](#) records the full provenance.

1 Introduction

All graphs are finite and simple. The *direct product* $G = \prod_{i=1}^n H_i$ of graphs $H_1, \dots, H_n$ has vertex set $V(H_1) \times \dots \times V(H_n)$, two vertices $(g_1, \dots, g_n)$ and $(g'_1, \dots, g'_n)$ being adjacent if and only if $g_i g'_i \in E(H_i)$ for every i . A graph is *complete multipartite* if its vertex set is partitioned into nonempty *parts* such that two vertices are adjacent exactly when they lie in different parts; following [1] we write $K[u, t]$ for the *balanced* complete multipartite graph with t parts of size u , so that $K[1, t] = K_t$.

For a graph G and $S \subseteq V(G)$ let $N[S]$ denote the closed neighbourhood of S , the union of S with all neighbours of its vertices. Following [1], S is *irredundant* if $N[S \setminus \{v\}] \neq N[S]$ for every $v \in S$; the *upper irredundance number* $\text{IR}(G)$ is the maximum size of an irredundant set, the *upper domination number* $\Gamma(G)$ is the maximum size of an irredundant dominating set, and $\alpha(G)$ is the independence number. Every maximal independent set is an irredundant dominating set, so

$$\alpha(G) \leq \Gamma(G) \leq \text{IR}(G) \tag{1}$$

for every graph G , the upper part of the domination chain. Unwinding the definition ([Observation 2.1](#)), S is irredundant if and only if every $v \in S$ has a *private neighbour* p with $N[p] \cap S = \{v\}$; taking $p = v$ shows that every independent set is irredundant.

The question of when equality $\alpha(G) = \text{IR}(G)$ holds in direct products of complete multipartite graphs arose from the study of unitary Cayley graphs. Defant and Iyer [3] conjectured that $\alpha(G) = \Gamma(G)$ whenever $G = \prod_{i=1}^n K[u_i, t_i]$ is a direct product of balanced complete multipartite graphs, and proved it when $\min_i t_i \leq 2$ or $n \leq 3$. Burcroff [2] observed that these proofs never use the domination

hypothesis, and conjectured the stronger statement $\alpha(G) = \text{IR}(G)$ for all such balanced products ([1, Conjecture 1.1]). Alon and Defant [1] proved Burcroff’s conjecture for every balanced product except 37 explicitly listed graphs, namely K_3^4 , $K[2, 3] \times K_3^3$, $K[3, 3] \times K_3^3$, $K_4 \times K_3^3$, ..., K_3^5 , K_3^6 and $K_4 \times K_3^5$ ([1, Theorem 1.2 and Section 3]), by combining an isoperimetric inequality, a stability theorem for large independent sets, and the following bound, proved there by the polynomial method:

Theorem 1.1 ([1, Theorem 1.3]). *Let $G = \prod_{i=1}^n H_i$, where each H_i is complete multipartite. If $S \subseteq V(G)$ is irredundant, then S is the disjoint union of an independent set $\text{Lon}(S)$ and a set $\text{Soc}(S)$ with $|\text{Soc}(S)| \leq 2^n$. In particular $\text{IR}(G) \leq \alpha(G) + 2^n$.*

At the end of their introduction they propose the unbalanced strengthening which is the subject of this note. We quote it verbatim from the problem statement supplied to the model [1].

Conjecture 1.2 ([1, Conjecture 1.2]). *If $H_1, \dots, H_n$ are complete multipartite graphs and $G = \prod_{i=1}^n H_i$, then $\alpha(G) = \text{IR}(G)$.*

The referee’s literature search (Appendix B) and the catalog’s earlier review found no published resolution of Conjecture 1.2; the source paper’s journal version [1] leaves it open. We prove it in full.

Theorem 1.3 (Main theorem). *Let $n \geq 1$ and let $H_1, \dots, H_n$ be complete multipartite graphs. Then*

$$\text{IR}\left(\prod_{i=1}^n H_i\right) = \alpha\left(\prod_{i=1}^n H_i\right).$$

Corollary 1.4. *Conjecture 1.2 holds. Consequently Burcroff’s conjecture [1, Conjecture 1.1] holds for all balanced products, including the 37 graphs left open in [1], and $\alpha(G) = \Gamma(G) = \text{IR}(G)$ for every direct product G of complete multipartite graphs; in particular the Defant–Iyer conjecture [3] holds.*

Proof. The first two statements are Theorem 1.3 and its special case. The third follows from Theorem 1.3 and the chain (1). $\square$

Idea of proof. Fix an irredundant set S and split it into its *lonely* vertices L (no neighbour in S) and its *t social* vertices, as in [1]. Vertices of G are labelled by the parts they occupy in each coordinate, and adjacency depends only on labels; the labels form the quotient graph $Q = \prod K_{q_i}$ with cell weights $w(x)$, and $\alpha(G)$ is the maximum weight $\alpha_w(Q)$ of an independent family of labels (Section 2). Encoding the label x as the tensor $z_x = \bigotimes_i (1, x_i) \in (\mathbb{R}^2)^{\otimes n}$, adjacency in Q becomes non-vanishing of the tensor product β of the 2×2 determinant forms. The Alon–Defant argument for Theorem 1.1 shows that the t social labels are linearly independent; the refinement here (Lemma 3.1) is that they stay independent modulo the span $M(\mathcal{A})$ of the lonely labels $\mathcal{A}$, and that everything lies in the β -orthogonal of $M(\mathcal{A})$, giving $t + 2\rho(\mathcal{A}) \leq 2^n$ with $\rho(\mathcal{A}) = \dim M(\mathcal{A})$. Complementing this, Lemma 3.3 shows that a nonempty independent label family of rank ρ misses the maximum weight by at least $2^n - 2\rho$: one compresses $\mathcal{A}$ coordinatewise to a down-set without decreasing weight or increasing rank (Lemma 3.2), identifies the rank of a down-set with the number of its $\{1, 2\}$ -points, hence with the size of an intersecting family $\mathcal{F} \subseteq 2^{[n]}$, and extends $\mathcal{F}$ to a maximal intersecting family of size 2^{n-1} , each added set contributing weight at least 2 because every factor has at least three parts. The two inequalities fit together exactly: $|S| \leq W(\mathcal{A}) + t \leq W(\mathcal{A}) + 2^n - 2\rho(\mathcal{A}) \leq \alpha_w(Q) = \alpha(G)$ whenever $L \neq \emptyset$. If $L = \emptyset$ then $|S| \leq 2^n$, which is at most $\alpha(G)$ for $n \geq 3$; the cases $n = 1$, $n = 2$ are settled by hand, the only delicate one being $K_3 \times K_3$. Products with a factor having at most two parts are edgeless or bipartite, and bipartite graphs satisfy $\text{IR} = \alpha$ (Lemma 2.2). [full proof]

2 The reduction and the weighted quotient

Observation 2.1. *For a graph G and $S \subseteq V(G)$, $N[S \setminus \{v\}] \neq N[S]$ holds if and only if there is a vertex p with $N[p] \cap S = \{v\}$. If v has a neighbour in S then any such p lies outside S and is adjacent to v .*

Proof. Since $N[S \setminus \{v\}] \subseteq N[S]$, inequality means some $p \in N[v]$ is not in $N[w]$ for any $w \in S \setminus \{v\}$; by symmetry of adjacency this says exactly $N[p] \cap S = \{v\}$. If $p \in S$ then $p \in N[p] \cap S = \{v\}$ forces $p = v$, whose closed neighbourhood meets S only in v , so v has no neighbour in S . $\square$

Let S be irredundant. Following [1] we call $v \in S$ *lonely* if it has no neighbour in S and *social* otherwise; write L for the set of lonely vertices and $t = |S \setminus L|$. By [Observation 2.1](#) every social vertex v has an *external private neighbour* $p_v \notin S$, adjacent to v and to no other vertex of S . Private neighbours chosen for distinct social vertices are distinct, since a common one would have two neighbours in S .

Lemma 2.2. *Every bipartite graph G satisfies $\text{IR}(G) = \alpha(G)$.*

Idea of proof. With bipartition $X \sqcup Y$ and S irredundant, replace each social vertex of $S \cap Y$ by its external private neighbour in X . The result is an independent set of the same size. [\[full proof\]](#)

If some factor H_i has no vertices, G is empty; if some H_i has one part, it is edgeless and so is G ; if some H_i has two parts, G is bipartite (colour a vertex by the part its i -th coordinate lies in). In all three cases $\text{IR}(G) = \alpha(G)$, the last by [Lemma 2.2](#). From now on every factor has $q_i \geq 3$ parts. Note that this reduction disposes of the whole product, not merely of the small factor: the lemmas below are stated for $q_i \geq 3$ only.

Labels, cells and weights. Write the parts of H_i as $P_{i,1}, \dots, P_{i,q_i}$ with sizes $s_{i,a} = |P_{i,a}|$, ordered so that $s_{i,1} \geq s_{i,2} \geq \dots \geq s_{i,q_i} \geq 1$. Put $\Omega = \prod_{i=1}^n [q_i]$. The *label* of a vertex v is $\lambda(v) \in \Omega$ with $\lambda(v)_i = a$ when $v_i \in P_{i,a}$; the *cell* of $x \in \Omega$ is $C_x = \lambda^{-1}(x) = \prod_i P_{i,x_i}$, of *weight* $w(x) = |C_x| = \prod_i s_{i,x_i} \geq 1$. Two vertices are adjacent in G if and only if their labels differ in every coordinate. Hence each cell is independent, and two distinct cells are either completely joined (labels differing everywhere) or completely non-adjacent. Call $x, y \in \Omega$ *adjacent* if they differ in every coordinate; this makes Ω the vertex set of $Q = \prod_i K_{q_i}$. A family $\mathcal{A} \subseteq \Omega$ is *independent* if any two distinct members agree in some coordinate; its weight is $W(\mathcal{A}) = \sum_{x \in \mathcal{A}} w(x)$, and

$$\alpha(G) = \alpha_w(Q) := \max\{W(\mathcal{J}) : \mathcal{J} \subseteq \Omega \text{ independent}\}, \quad (2)$$

because an independent set of G may be enlarged to the union of the cells it meets, and conversely $\bigcup_{x \in \mathcal{J}} C_x$ is independent of size $W(\mathcal{J})$ for every independent $\mathcal{J}$.

The tensor model. For each coordinate i and label $a \in [q_i]$ put $u_{i,a} = (1, a) \in \mathbb{R}^2$, and for $x \in \Omega$ put

$$z_x = \bigotimes_{i=1}^n u_{i,x_i} \in V := (\mathbb{R}^2)^{\otimes n}, \quad \dim V = 2^n.$$

For $\mathcal{A} \subseteq \Omega$ let $M(\mathcal{A}) = \text{span}\{z_x : x \in \mathcal{A}\}$ and $\rho(\mathcal{A}) = \dim M(\mathcal{A})$. Equip V with the bilinear form β which is the tensor product of the determinant forms,

$$\beta\left(\bigotimes_i v_i, \bigotimes_i v'_i\right) = \prod_{i=1}^n \det(v_i, v'_i), \quad \text{so that} \quad \beta(z_x, z_y) = \prod_{i=1}^n (y_i - x_i).$$

Thus $\beta(z_x, z_y) \neq 0$ if and only if x and y are adjacent in Q , and $\beta(z_x, z_x) = 0$. The form is non-degenerate, its Gram matrix in the basis $\{e_{b_1} \otimes \dots \otimes e_{b_n}\}$ being the tensor power $J^{\otimes n}$ of the invertible matrix $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$; it is symmetric for even n and alternating for odd n , and in either case $\beta(v, v') = 0$ iff $\beta(v', v) = 0$ and $\dim U^\perp = 2^n - \dim U$ for every subspace $U \leq V$, where $U^\perp = \{v : \beta(v, u) = 0 \ \forall u \in U\}$.

Remark 2.3 (Labelling dependence). The rank $\rho(\mathcal{A})$ depends on the identification of the parts of H_i with $[q_i]$, and [Lemma 3.2\(b\)](#) uses the decreasing-size ordering fixed above. This is harmless because all statements below refer to one fixed labelling; the referee asked that this be said explicitly ([Appendix B](#)).

3 The two inequalities

Lemma 3.1 (Rank-refined social bound). *Let $q_i \geq 3$ for all i , let $S \subseteq V(G)$ be irredundant with lonely set L and t social vertices, and let $\mathcal{A} = \lambda(L) \subseteq \Omega$. Then*

$$t + 2\rho(\mathcal{A}) \leq 2^n. \quad (3)$$

In particular $t \leq 2^n$, which is Theorem 1.1 of Alon and Defant; and distinct social vertices lie in distinct cells.

Idea of proof. Let $x_1, \dots, x_t$ be the labels of the social vertices and $y_1, \dots, y_t$ those of their external private neighbours. Privacy says $\beta(z_{y_j}, z_{x_k}) \neq 0$ iff $j = k$, and every z_{y_j} is β -orthogonal to $M(\mathcal{A})$, since a private neighbour is non-adjacent to every lonely vertex. Pairing a relation $\sum_k c_k z_{x_k} \in M(\mathcal{A})$ with z_{y_j} gives $c_j = 0$, so $\dim(M(\mathcal{A}) + \text{span}\{z_{x_k}\}) = \rho(\mathcal{A}) + t$. On the other hand this whole space lies in $M(\mathcal{A})^\perp$: lonely vertices are pairwise non-adjacent and non-adjacent to social ones. Nondegeneracy gives $\rho(\mathcal{A}) + t \leq 2^n - \rho(\mathcal{A})$. The case $\mathcal{A} = \emptyset$ is the linear-independence argument of [1, proof of Theorem 1.3], there phrased with the multilinear polynomials $f_v = \prod_i (x_i - c_v(i))$, which are the linear functionals $\beta(z_{\lambda(v)}, \cdot)$ of the present model. [\[full proof\]](#)

For a coordinate i write $\Omega_{-i} = \prod_{j \neq i} [q_j]$, and for $\hat{x} \in \Omega_{-i}$ and $a \in [q_i]$ write $(\hat{x}, a) \in \Omega$ for the label with i -th coordinate a and remaining coordinates $\hat{x}$. The fibre of $\mathcal{A} \subseteq \Omega$ over $\hat{x}$ is $A_{\hat{x}} = \{a \in [q_i] : (\hat{x}, a) \in \mathcal{A}\}$. The compression $C_i \mathcal{A}$ replaces each fibre by an initial segment of the same size: $(C_i \mathcal{A})_{\hat{x}} = \{1, \dots, |A_{\hat{x}}|\}$. Clearly $|C_i \mathcal{A}| = |\mathcal{A}|$.

Lemma 3.2 (Compression). *For every $\mathcal{A} \subseteq \Omega$ and every coordinate i :*

- (a) *if $\mathcal{A}$ is independent then so is $C_i \mathcal{A}$;*
- (b) *$W(C_i \mathcal{A}) \geq W(\mathcal{A})$;*
- (c) *$\rho(C_i \mathcal{A}) \leq \rho(\mathcal{A})$.*

Consequently, repeatedly compressing coordinates that change the family terminates in a down-set $\mathcal{D}$ (closed under coordinatewise decrease) with $|\mathcal{D}| = |\mathcal{A}|$, $W(\mathcal{D}) \geq W(\mathcal{A})$, $\rho(\mathcal{D}) \leq \rho(\mathcal{A})$, and $\mathcal{D}$ independent if $\mathcal{A}$ is.

Idea of proof. (a): two members of $C_i \mathcal{A}$ whose other coordinates differ everywhere come from two fibres each of which, by independence of $\mathcal{A}$, is the same singleton; both are sent to 1. (b): a fibre of size k contributes $\sum_{a \in A_{\hat{x}}} s_{i,a} \leq s_{i,1} + \dots + s_{i,k}$ in coordinate i , by the decreasing order of part sizes. (c): write $V = \mathbb{R}^2 \otimes V_{-i}$ and let T, E be the spans in V_{-i} of the tensors of the rest-tuples with nonempty, respectively at least two-element, fibres. Projecting along $\ell(a, b) = a$ maps $M(\mathcal{A})$ onto T , while differences within a fibre put $(0, 1) \otimes E$ into $M(\mathcal{A}) \cap \ker(\ell \otimes \text{id})$, so $\rho(\mathcal{A}) \geq \dim T + \dim E$ by rank-nullity; after compression $M(C_i \mathcal{A}) = u_{i,1} \otimes T \oplus (0, 1) \otimes E$ has dimension exactly $\dim T + \dim E$. Termination: each effective compression strictly decreases $\sum_{x \in \mathcal{A}} \sum_i x_i$. [\[full proof\]](#)

Lemma 3.3 (Weighted rank-deficit inequality). *Let $q_i \geq 3$ for all i . For every nonempty independent family $\mathcal{A} \subseteq \Omega$,*

$$\alpha_w(Q) - W(\mathcal{A}) \geq 2^n - 2\rho(\mathcal{A}). \quad (4)$$

Idea of proof. Compress $\mathcal{A}$ to a down-set $\mathcal{D}$. Since $\{z_b : b \in \{1, 2\}^n\}$ is a basis of V and $u_{i,a} = (2 - a)u_{i,1} + (a - 1)u_{i,2}$, every z_x with $x \in \mathcal{D}$ expands over z_b with $b \leq x$, hence $b \in \mathcal{D}$; so $\rho(\mathcal{D}) = |\mathcal{D} \cap \{1, 2\}^n|$. Encoding $T \subseteq [n]$ by $b(T) \in \{1, 2\}^n$ with $b(T)_i = 1$ iff $i \in T$, the family $\mathcal{F} = \{T : b(T) \in \mathcal{D}\}$ has $|\mathcal{F}| = \rho(\mathcal{D})$, is upward closed, contains $[n]$, and is intersecting (two disjoint members would give two members of $\mathcal{D}$ differing everywhere). For an intersecting $\mathcal{H}$ the family $E(\mathcal{H}) = \{x \in \Omega : \{i : x_i = 1\} \in \mathcal{H}\}$ is independent, and $\mathcal{D} \subseteq E(\mathcal{F})$. Extend $\mathcal{F}$

to a maximal intersecting family $\mathcal{F}^*$, of size 2^{n-1} (it contains exactly one of each complementary pair). Each $T \in \mathcal{F}^* \setminus \mathcal{F}$ misses some coordinate i , so the labels with $\{i : x_i = 1\} = T$ have total weight $m(T) = \prod_{i \in T} s_{i,1} \prod_{i \notin T} \sum_{a \geq 2} s_{i,a} \geq q_i - 1 \geq 2$. Hence $\alpha_w(Q) - W(\mathcal{A}) \geq W(E(\mathcal{F}^*)) - W(E(\mathcal{F})) \geq 2(2^{n-1} - |\mathcal{F}|) = 2^n - 2\rho(\mathcal{D}) \geq 2^n - 2\rho(\mathcal{A})$. *[full proof]*

Remark 3.4 (Tightness). Both inequalities are sharp. A full coordinate fibre $\mathcal{J} = \{x : x_1 = 1\}$ has $\rho(\mathcal{J}) = 2^{n-1}$ and $W(\mathcal{J}) = \alpha_w(Q)$ when $s_{1,1}$ is the largest part size overall, so (4) holds with equality, and for $S = \bigcup_{x \in \mathcal{J}} C_x$ (all lonely, $t = 0$) (3) reads $2 \cdot 2^{n-1} \leq 2^n$. The referee’s exhaustive sweeps found slack exactly 0 at these families and nowhere smaller (*Remark 4.4*).

4 Remarks

Remark 4.1 (What is new relative to [1]). The $\rho = 0$ case of *Lemma 3.1* is exactly *Theorem 1.1*, and its proof is the Alon–Defant polynomial-method proof transposed into tensor language; it is *not* new and is attributed accordingly (the original writeup proved it afresh without citation, a defect noted by the referee). The new ingredients are the term $2\rho(\mathcal{A})$ in (3), which retains information about the lonely vertices instead of discarding them, and the deficit inequality (4) with its compression proof, whose lower bound $2^n - 2\rho(\mathcal{A})$ is precisely what (3) needs. The argument uses neither the isoperimetric inequality nor the stability theorem of [1], and it makes no use of balance, so it applies verbatim to the 37 balanced products listed in [1, Section 3].

Remark 4.2 (Scope of the hypotheses). *Lemmas 3.1* and *3.3* are stated for $q_i \geq 3$; the inequality (4) uses $q_i \geq 3$ exactly once, in the bound $m(T) \geq 2$. Products with a factor having at most two parts are handled globally by *Lemma 2.2*, and one cannot simply delete such a factor and apply the lemmas to the rest. The degenerate cases $n = 0$, an empty factor, and $n = 1, 2$ are all covered: in the lonely-free case $n \geq 3$ uses $3^{n-1} \geq 2^n$; $n = 1$ uses that a complete graph has no irredundant set of size 2; and $n = 2$ reduces to the single graph $K_3 \times K_3$, whose 4-element all-social irredundant sets are excluded by a short case analysis in *Appendix A* (brute force gives $\text{IR}(K_3 \times K_3) = 3 = \alpha$). At the level of labels alone, a configuration with $t = 4 = 2^2$ social labels and $\mathcal{A} = \emptyset$ does exist in $K_3 \times K_3$; it is excluded only because the required private neighbour would have to be a selected vertex. This is why $K_3 \times K_3$ needs a separate argument.

Remark 4.3 (Relation to the earlier attempt). This is a retry target (*Section 5*). The first attempt, by gpt-5.6-so1, established *Conjecture 1.2* for $n \leq 3$, for products with a factor having at most two parts, and under the arithmetic condition $\prod_i \gcd_a s_{i,a} \geq 2^n$, via a “blocker inequality” $W(\mathcal{A}) + W(B(\mathcal{A})) \leq 2\alpha_w$ valid in three coordinates; it recorded as an obstruction that this inequality fails for $\mathcal{A} = \{(0, 0, 0, 0), (1, 0, 0, 0)\}$ in K_3^4 , where $|\mathcal{A}| + |B(\mathcal{A})| = 59 > 54$. The present argument does not use blockers, and the referee checked that the obstruction is irrelevant to it: that family has $\rho(\mathcal{A}) = 2$, and (4) asks only $27 - 2 \geq 16 - 4$. What the two attempts share is the quotient reduction, the bipartite lemma (*Lemma 2.2*, essentially the same proof in both) and the tensor formulation of the bound $t \leq 2^n$; these are inherited and, as said above, the last is due to [1].

Remark 4.4 (The referee’s computational checks). The adversarial referee (*Appendix B*; scripts in `verification_astra/scripts/1904.02595__00/`) computed IR by exhaustive search over irredundant sets, which is complete because irredundance is hereditary, and α by maximum-weight clique in the complement of Q . It found $\text{IR} = \alpha$ in 76 products with at most 18 vertices (one to three factors, balanced and unbalanced) and in four larger exhaustive runs: K_3^3 (27 vertices, 207,244 irredundant sets, $\alpha = \text{IR} = 9$), $K[2, 1, 1] \times K_3 \times K_2$ (24 vertices, 140,016 sets, $\alpha = \text{IR} = 12$), $K[2, 2] \times K_3 \times K_3$ (36 vertices, 4,223,278 sets, $\alpha = \text{IR} = 18$) and $K[2, 1, 1] \times K_3 \times K_3$ (36 vertices, 2,020,728 sets, $\alpha = \text{IR} = 18$); here $K[2, 1, 1]$ denotes the complete multipartite graph with parts of sizes 2, 1, 1. Inequality (3) was verified, with exact rational ranks, over all 207,244 irredundant sets of K_3^3 and over adversarially built label configurations in K_3^2 , K_3^3 and K_3^4 , with minimum slack 0. *Lemma 3.2* was checked on 300 random families and the identity $\rho(\mathcal{D}) = |\mathcal{D} \cap \{1, 2\}^n|$ on 400 random down-sets, with

no failure. Inequality (4) was checked exhaustively over all independent families for $n \leq 2$ ($q_i \leq 5$), over all families of size ≤ 3 for $n = 3$ and ≤ 2 for $n = 4$, and over random maximal families, with unit and random weights: no violation, minimum slack 0 attained at maximum independent families with $\rho = 2^{n-1}$. For two of the 37 open cases of [1], K_3^4 and K_3^5 , a local search for $\max(|\mathcal{A}| - 2\rho(\mathcal{A}))$ returned exactly $\alpha - 2^n = 11$ and 49, so (4) is tight there; a seeded search for a large irredundant set in K_3^4 found nothing beyond $|S| = 27 = \alpha$.

Remark 4.5 (Status). The referee could not find an error and every load-bearing inequality survived exhaustive attack, including inside the exceptional cases of [1]. It nevertheless stressed, and we repeat, that a three-page argument superseding the main theorem of a published paper of Alon and Defant should receive independent human verification before being relied upon. Nothing in this note has been checked by a human mathematician.

5 Provenance

The mathematics in this note was produced without human intervention, in three stages.

(1) The argument was found by the language model gpt-6-astra (reasoning effort *max*, single autonomous pass, flex service tier) on 2026-09-16, as part of a sweep over open problems catalogued from arXiv (catalog id 1904.02595__00, leg attacks_retry; original writeup in attacks_retry/1904.02595__00/output.md). This was a *second* attempt at the problem: the first, by gpt-5.6-sol (attacks/1904.02595__00, self-reported verdict *partial*), was supplied verbatim in the prompt as an unverified lead, without a referee report on it. Its content is summarised in Remark 4.3. The second writeup inherits from the first the quotient reduction, the bipartite lemma and the tensor formulation of the bound $t \leq 2^n$, and states that it reproves everything it uses; its new steps are the rank refinement (3), the compression lemma and the deficit inequality (4), and the lonely-free case analysis. Nothing inherited is presented here as new.

(2) An independent adversarial referee agent (claude-opus-5) reviewed the writeup on 2026-09-17. It fetched [1], confirmed the definitions of irredundance and IR, the statements of Conjectures 1.1 and 1.2 and of Theorem 1.3 and the list of 37 exceptional graphs; verified that the writeup’s private-neighbour definition is equivalent to the source’s; re-derived all 21 numbered steps and rated each *VALID*; ran the computational checks summarised in Remark 4.4; searched for prior or concurrent resolutions and found none; and returned the verdict *CONFIRMED* with high confidence. Its only findings were expository (the implicit “distinct social vertices lie in distinct cells” and the labelling dependence of ρ) and one of scholarship (the $\rho = 0$ case of the rank bound is Theorem 1.3 of [1] and must be cited). Its report is reproduced verbatim in Appendix B.

(3) On 2026-09-17 the writeup was rewritten into the present note by claude-fable-5-1. The text was reorganised, with statements and proof ideas in the main text and full proofs in Appendix A. Every deviation from the original writeup is listed here.

- Theorem 1.1 is stated and cited, and Lemma 3.1 is presented as its refinement rather than as a new result; the relation between the polynomial-method proof of [1] and the tensor model is stated in the proof idea and in Remark 4.1. (Scholarship repair named by the referee.)
- The statement that distinct social vertices occupy distinct cells, implicit in the writeup’s enumeration of social labels, is made explicit in Lemma 3.1 and proved in Appendix A. (Clarification named by the referee.)
- Remark 2.3 on the labelling dependence of ρ and the load-bearing role of the decreasing-size ordering was added. (Clarification named by the referee.)
- Observation 2.1, proving the equivalence of the source’s definition of irredundance with the private-neighbour form used in the writeup, was added from the referee’s interpretation check.

- The historical context (Defant–Iyer [3], Burcroff [2], the 37 cases of [1]) and [Corollary 1.4](#), which records the consequences for Burcroff’s and the Defant–Iyer conjectures noted by the referee, were added; the writeup stated only [Conjecture 1.2](#).
- In the proof of [Lemma 3.2\(a\)](#), the case in which the two compressed labels have identical remaining coordinates is spelled out; in (c), the directness of the sum $u_{i,1} \otimes T \oplus (0, 1) \otimes E$ is justified; in [Lemma 3.3](#), the extension of $\mathcal{F}$ to a maximal intersecting family and the inequality chain in the last display are written out in full. These are the writeup’s arguments with the steps the referee verified made explicit.
- [Remarks 3.4](#) and [4.2](#) to [4.5](#) were added from the referee report.

No other mathematical content was changed. **No human mathematician has checked this argument.** We circulate it for human review, not as a claim to a resolution.

A Full proofs

Throughout, $G = \prod_{i=1}^n H_i$ with H_i complete multipartite, and the notation of [Section 2](#) is in force.

Lemma 2.2. *Every bipartite graph G satisfies $\text{IR}(G) = \alpha(G)$.*

Proof. Let $V(G) = X \sqcup Y$ be a bipartition and let S be irredundant. Let L_Y be the set of lonely vertices of $S \cap Y$, and for each social $y \in S \cap Y$ choose an external private neighbour p_y ; as p_y is adjacent to $y \in Y$, $p_y \in X \setminus S$. Put

$$I = (S \cap X) \cup L_Y \cup \{p_y : y \in S \cap Y \text{ social}\}.$$

We claim I is independent. The vertices of $S \cap X$ and the p_y all lie in X , so no edge joins two of them. A vertex of L_Y has no neighbour in S , in particular none in $S \cap X$, and two vertices of $L_Y \subseteq Y$ are non-adjacent. Finally p_y is non-adjacent to every vertex of S other than y , in particular to every vertex of L_Y (note $y \notin L_Y$ as y is social). The private neighbours p_y are pairwise distinct and lie outside S , so

$$|I| = |S \cap X| + |L_Y| + |\{y \in S \cap Y \text{ social}\}| = |S \cap X| + |S \cap Y| = |S|.$$

Hence $|S| \leq \alpha(G)$, and $\text{IR}(G) \leq \alpha(G)$. The reverse inequality holds since every independent set is irredundant. $\square$

Lemma 3.1 (Rank-refined social bound). *Let $q_i \geq 3$ for all i , let $S \subseteq V(G)$ be irredundant with lonely set L and t social vertices, and let $\mathcal{A} = \lambda(L) \subseteq \Omega$. Then*

$$t + 2\rho(\mathcal{A}) \leq 2^n. \tag{3}$$

In particular $t \leq 2^n$, which is [Theorem 1.1](#) of Alon and Defant; and distinct social vertices lie in distinct cells.

Proof. Enumerate the social vertices as $v_1, \dots, v_t$ with labels $x_k = \lambda(v_k)$, and choose external private neighbours $p_1, \dots, p_t$ with labels $y_k = \lambda(p_k)$. Since adjacency in G is adjacency of labels in Q , and $\beta(z_x, z_y) \neq 0$ iff x, y are adjacent in Q :

- (i) $\beta(z_{y_j}, z_{x_k}) \neq 0$ if $j = k$, because p_j is adjacent to v_j ; and $\beta(z_{y_j}, z_{x_k}) = 0$ if $j \neq k$, because p_j is not adjacent to $v_k \in S \setminus \{v_j\}$.
- (ii) $\beta(z_{y_j}, z_x) = 0$ for every $x \in \mathcal{A}$: a lonely vertex in C_x belongs to $S \setminus \{v_j\}$, so it is not adjacent to p_j . Hence $\beta(z_{y_j}, m) = 0$ for all $m \in M(\mathcal{A})$.

- (iii) $\beta(z_x, z_{x'}) = 0$ for all $x, x' \in \mathcal{A}$: for $x = x'$ every factor $x'_i - x_i$ vanishes, and for $x \neq x'$ two lonely vertices in $C_x, C_{x'}$ are non-adjacent, so the labels agree in some coordinate. Thus $M(\mathcal{A}) \subseteq M(\mathcal{A})^\perp$.
- (iv) $\beta(z_{x_k}, z_x) = 0$ for all k and $x \in \mathcal{A}$: a social vertex adjacent to a lonely vertex would make the latter non-lonely. Thus $z_{x_k} \in M(\mathcal{A})^\perp$.

By (i), distinct social vertices have distinct labels: if $x_j = x_k$ with $j \neq k$ then $\beta(z_{y_j}, z_{x_k}) = \beta(z_{y_j}, z_{x_j})$ would be both zero and nonzero. (Equivalently: a private neighbour of one of two vertices in the same cell would be adjacent to the other.)

The vectors $z_{x_1}, \dots, z_{x_t}$ are linearly independent modulo $M(\mathcal{A})$. Indeed, suppose $\sum_{k=1}^t c_k z_{x_k} \in M(\mathcal{A})$. Applying $\beta(z_{y_j}, \cdot)$ and using (ii) and (i),

$$0 = \beta\left(z_{y_j}, \sum_k c_k z_{x_k}\right) = \sum_k c_k \beta(z_{y_j}, z_{x_k}) = c_j \beta(z_{y_j}, z_{x_j}),$$

so $c_j = 0$ for every j . Therefore

$$\dim(M(\mathcal{A}) + \text{span}\{z_{x_1}, \dots, z_{x_t}\}) = \rho(\mathcal{A}) + t. \quad (5)$$

By (iii) and (iv) the subspace in (5) is contained in $M(\mathcal{A})^\perp$, and by nondegeneracy of β , $\dim M(\mathcal{A})^\perp = 2^n - \rho(\mathcal{A})$. Hence $\rho(\mathcal{A}) + t \leq 2^n - \rho(\mathcal{A})$, which is (3). When $\mathcal{A} = \emptyset$ this reads $t \leq 2^n$, the bound of Theorem 1.1. $\square$

Lemma 3.2 (Compression). *For every $\mathcal{A} \subseteq \Omega$ and every coordinate i :*

- (a) *if $\mathcal{A}$ is independent then so is $C_i \mathcal{A}$;*
- (b) $W(C_i \mathcal{A}) \geq W(\mathcal{A})$;
- (c) $\rho(C_i \mathcal{A}) \leq \rho(\mathcal{A})$.

Consequently, repeatedly compressing coordinates that change the family terminates in a down-set $\mathcal{D}$ (closed under coordinatewise decrease) with $|\mathcal{D}| = |\mathcal{A}|$, $W(\mathcal{D}) \geq W(\mathcal{A})$, $\rho(\mathcal{D}) \leq \rho(\mathcal{A})$, and $\mathcal{D}$ independent if $\mathcal{A}$ is.

Proof. Fix the coordinate i and write labels as $(\hat{x}, a)$ with $\hat{x} \in \Omega_{-i}$, $a \in [q_i]$.

(a) Let $(\hat{x}, a) \neq (\hat{y}, a')$ be two members of $C_i \mathcal{A}$. If $\hat{x}$ and $\hat{y}$ agree in some coordinate $j \neq i$, the two labels agree there and we are done. Otherwise $\hat{x}$ and $\hat{y}$ differ in every coordinate $j \neq i$ (if $n = 1$ this is vacuous and $\hat{x} = \hat{y}$). Both original fibres $A_{\hat{x}}, A_{\hat{y}}$ are nonempty. For $b \in A_{\hat{x}}$ and $b' \in A_{\hat{y}}$, the labels $(\hat{x}, b), (\hat{y}, b') \in \mathcal{A}$ must agree in some coordinate, which can only be i , so $b = b'$. Hence $A_{\hat{x}} = A_{\hat{y}} = \{b\}$ for a single b (if $\hat{x} = \hat{y}$ this says the fibre is a singleton). Compression sends both fibres to $\{1\}$, so $a = a' = 1$; the two labels agree in coordinate i (and if $\hat{x} = \hat{y}$ they coincide, contrary to assumption). Thus $C_i \mathcal{A}$ is independent.

(b) Grouping by rest-tuples,

$$W(\mathcal{A}) = \sum_{\hat{x} \in \Omega_{-i}} \left(\prod_{j \neq i} s_{j, \hat{x}_j} \right) \sum_{a \in A_{\hat{x}}} s_{i, a},$$

and for a fibre of size k , $\sum_{a \in A_{\hat{x}}} s_{i, a} \leq s_{i, 1} + \dots + s_{i, k}$, because $s_{i, 1} \geq \dots \geq s_{i, q_i}$ and the right side is the sum of the k largest part sizes. The right side is the contribution of the compressed fibre, so $W(C_i \mathcal{A}) \geq W(\mathcal{A})$.

(c) Identify $V = \mathbb{R}^2 \otimes V_{-i}$ with $V_{-i} = \bigotimes_{j \neq i} \mathbb{R}^2$, so that $z_{(\hat{x}, a)} = u_{i, a} \otimes \hat{z}_{\hat{x}}$ with $\hat{z}_{\hat{x}} = \bigotimes_{j \neq i} u_{j, \hat{x}_j}$. Put

$$T = \text{span}\{\hat{z}_{\hat{x}} : A_{\hat{x}} \neq \emptyset\}, \quad E = \text{span}\{\hat{z}_{\hat{x}} : |A_{\hat{x}}| \geq 2\} \leq T,$$

let $M = M(\mathcal{A})$, let $\ell : \mathbb{R}^2 \rightarrow \mathbb{R}$, $\ell(a, b) = a$, and $\pi = \ell \otimes \text{id} : V \rightarrow V_{-i}$. Since $\ell(u_{i,a}) = 1$ for every a , $\pi(z_{(\hat{x},a)}) = \hat{z}_{\hat{x}}$, whence $\pi(M) = T$. If $|A_{\hat{x}}| \geq 2$, pick $a \neq a'$ in $A_{\hat{x}}$; then $z_{(\hat{x},a)} - z_{(\hat{x},a')} = (u_{i,a} - u_{i,a'}) \otimes \hat{z}_{\hat{x}} = (0, a - a') \otimes \hat{z}_{\hat{x}} \in M$, so $e_2 \otimes \hat{z}_{\hat{x}} \in M$ with $e_2 = (0, 1)$. Hence $e_2 \otimes E \subseteq M$, and $e_2 \otimes E \subseteq \ker \pi$ as $\ell(e_2) = 0$. Rank-nullity for $\pi|_M$ gives

$$\rho(\mathcal{A}) = \dim M = \dim \pi(M) + \dim(M \cap \ker \pi) \geq \dim T + \dim(e_2 \otimes E) = \dim T + \dim E,$$

the map $\hat{z} \mapsto e_2 \otimes \hat{z}$ being injective. After compression, a rest-tuple with a fibre of size 1 contributes $u_{i,1} \otimes \hat{z}_{\hat{x}}$, and one with a fibre of size $k \geq 2$ contributes $u_{i,1} \otimes \hat{z}_{\hat{x}}, \dots, u_{i,k} \otimes \hat{z}_{\hat{x}}$, whose span is $\text{span}\{u_{i,1}, u_{i,2}\} \otimes \hat{z}_{\hat{x}} = \text{span}\{u_{i,1}, e_2\} \otimes \hat{z}_{\hat{x}}$ since $u_{i,2} - u_{i,1} = e_2$ and $\{u_{i,1}, u_{i,2}\}$ spans $\mathbb{R}^2$. Therefore

$$M(C_i \mathcal{A}) = (u_{i,1} \otimes T) + (e_2 \otimes E),$$

and the sum is direct: π restricted to $u_{i,1} \otimes V_{-i}$ is the injective map $u_{i,1} \otimes \hat{z} \mapsto \hat{z}$, while π vanishes on $e_2 \otimes V_{-i}$. So $\rho(C_i \mathcal{A}) = \dim T + \dim E \leq \rho(\mathcal{A})$.

Termination and the down-set. Compression maps a fibre $\{a_1 < \dots < a_k\}$ to $\{1, \dots, k\}$ with $a_j \geq j$; if the fibre changes then $\sum_j a_j > \sum_j j$. Hence every compression that changes the family strictly decreases the positive integer $\sum_{x \in \mathcal{A}} \sum_i x_i$, and the process stops after finitely many steps at a family $\mathcal{D}$ all of whose fibres, in every coordinate, are initial segments. Such a family is a down-set: if $x \in \mathcal{D}$ and $y \leq x$ coordinatewise, decreasing one coordinate at a time stays within the initial-segment fibres, so $y \in \mathcal{D}$. Compression preserves cardinality (it is a bijection on each fibre and preserves rest-tuples), and the three properties (a)–(c) persist along the iteration. $\square$

Lemma 3.3 (Weighted rank-deficit inequality). *Let $q_i \geq 3$ for all i . For every nonempty independent family $\mathcal{A} \subseteq \Omega$,*

$$\alpha_w(Q) - W(\mathcal{A}) \geq 2^n - 2\rho(\mathcal{A}). \quad (4)$$

Proof. By Lemma 3.2, compress $\mathcal{A}$ to a down-set $\mathcal{D}$, which is nonempty and independent with $W(\mathcal{D}) \geq W(\mathcal{A})$ and $\rho(\mathcal{D}) \leq \rho(\mathcal{A})$. It suffices to prove $\alpha_w(Q) - W(\mathcal{D}) \geq 2^n - 2\rho(\mathcal{D})$.

Step 1: $\rho(\mathcal{D}) = |\mathcal{D} \cap \{1, 2\}^n|$. Since $u_{i,1} = (1, 1)$ and $u_{i,2} = (1, 2)$ form a basis of $\mathbb{R}^2$, the tensors z_b , $b \in \{1, 2\}^n$, form a basis of V . For $a \in [q_i]$, $u_{i,a} = (2-a)u_{i,1} + (a-1)u_{i,2}$; when $a = 1$ only $u_{i,1}$ occurs, and when $a \geq 2$ both $u_{i,1}, u_{i,2}$ have i -th label $\leq a$. Expanding the tensor product, z_x is a linear combination of the z_b with $b \in \{1, 2\}^n$ and $b \leq x$ coordinatewise. For $x \in \mathcal{D}$ all such b lie in $\mathcal{D}$, because $\mathcal{D}$ is a down-set. Hence $M(\mathcal{D}) = \text{span}\{z_b : b \in \mathcal{D} \cap \{1, 2\}^n\}$, and these z_b are linearly independent as a subset of a basis.

Step 2: the intersecting family. For $T \subseteq [n]$ let $b(T) \in \{1, 2\}^n$ have $b(T)_i = 1$ for $i \in T$ and $b(T)_i = 2$ for $i \notin T$, and put $\mathcal{F} = \{T \subseteq [n] : b(T) \in \mathcal{D}\}$. By Step 1, $|\mathcal{F}| = \rho(\mathcal{D})$. If $T \subseteq U$ then $b(U) \leq b(T)$ coordinatewise, so $\mathcal{F}$ is upward closed. As $\mathcal{D}$ is a nonempty down-set it contains $(1, \dots, 1) = b([n])$, so $[n] \in \mathcal{F}$. Finally $\mathcal{F}$ is intersecting: if $T, U \in \mathcal{F}$ were disjoint then $T \subseteq [n] \setminus U$, so $[n] \setminus U \in \mathcal{F}$ by upward closure, and $b(U), b([n] \setminus U)$ would be two members of $\mathcal{D}$ differing in every coordinate, contradicting independence of $\mathcal{D}$.

Step 3: pattern classes. For an intersecting family $\mathcal{H} \subseteq 2^{[n]}$ put $E(\mathcal{H}) = \{x \in \Omega : \{i : x_i = 1\} \in \mathcal{H}\}$. Two members of $E(\mathcal{H})$ have 1-patterns in $\mathcal{H}$, which intersect, so they share a coordinate equal to 1; thus $E(\mathcal{H})$ is independent. Moreover $\mathcal{D} \subseteq E(\mathcal{F})$: for $x \in \mathcal{D}$ let $T = \{i : x_i = 1\}$; then $b(T) \leq x$ coordinatewise ($b(T)_i = 1 = x_i$ on T , and $b(T)_i = 2 \leq x_i$ off T), so $b(T) \in \mathcal{D}$ and $T \in \mathcal{F}$. For $T \subseteq [n]$ the labels with 1-pattern exactly T have total weight

$$m(T) = \prod_{i \in T} s_{i,1} \prod_{i \notin T} \left(\sum_{a=2}^{q_i} s_{i,a} \right), \quad \text{so} \quad W(E(\mathcal{H})) = \sum_{T \in \mathcal{H}} m(T).$$

Step 4: maximal intersecting families. Since $\mathcal{F}$ is intersecting and $2^{[n]}$ is finite, $\mathcal{F}$ is contained in an inclusion-maximal intersecting family $\mathcal{F}^*$. We use the well-known fact $|\mathcal{F}^*| = 2^{n-1}$, with its short

proof. A maximal intersecting family is upward closed: if $T \in \mathcal{F}^*$ and $U \supseteq T$ then U meets every member of $\mathcal{F}^*$, so $\mathcal{F}^* \cup \{U\}$ is intersecting and $U \in \mathcal{F}^*$ by maximality. If $T \notin \mathcal{F}^*$, maximality gives $U \in \mathcal{F}^*$ with $T \cap U = \emptyset$, i.e. $U \subseteq [n] \setminus T$, whence $[n] \setminus T \in \mathcal{F}^*$. An intersecting family contains at most one of T and $[n] \setminus T$. So $\mathcal{F}^*$ contains exactly one set from each of the 2^{n-1} complementary pairs.

Step 5: the bound. Let $T \in \mathcal{F}^* \setminus \mathcal{F}$. Since $[n] \in \mathcal{F}$, $T \neq [n]$, so some $i \notin T$; the corresponding factor of $m(T)$ is $\sum_{a=2}^{q_i} s_{i,a} \geq q_i - 1 \geq 2$, and all other factors are at least 1. Thus $m(T) \geq 2$, and

$$W(E(\mathcal{F}^*)) - W(E(\mathcal{F})) = \sum_{T \in \mathcal{F}^* \setminus \mathcal{F}} m(T) \geq 2(2^{n-1} - |\mathcal{F}|) = 2^n - 2\rho(\mathcal{D}).$$

Since $E(\mathcal{F}^*)$ is independent, $\alpha_w(Q) \geq W(E(\mathcal{F}^*))$; since $\mathcal{D} \subseteq E(\mathcal{F})$ and weights are positive, $W(E(\mathcal{F})) \geq W(\mathcal{D})$. Therefore

$$\alpha_w(Q) - W(\mathcal{D}) \geq W(E(\mathcal{F}^*)) - W(E(\mathcal{F})) \geq 2^n - 2\rho(\mathcal{D}) \geq 2^n - 2\rho(\mathcal{A}),$$

and combining with $W(\mathcal{D}) \geq W(\mathcal{A})$ gives (4). $\square$

Theorem 1.3 (Main theorem). *Let $n \geq 1$ and let $H_1, \dots, H_n$ be complete multipartite graphs. Then*

$$\text{IR}\left(\prod_{i=1}^n H_i\right) = \alpha\left(\prod_{i=1}^n H_i\right).$$

Proof. Every independent set is irredundant, so $\alpha(G) \leq \text{IR}(G)$ and it suffices to show $|S| \leq \alpha(G)$ for every irredundant S . By the reduction of Section 2 (empty factor; a factor with one part makes G edgeless, in which case every set is independent; a factor with two parts makes G bipartite, and Lemma 2.2 applies) we may assume $q_i \geq 3$ for all i . Let S be irredundant with lonely set L , t social vertices, and $\mathcal{A} = \lambda(L)$.

Case 1: $L \neq \emptyset$. Then $\mathcal{A}$ is nonempty, and independent in Q : two lonely vertices in distinct cells are non-adjacent, so their labels agree somewhere. As $L \subseteq \bigcup_{x \in \mathcal{A}} C_x$, $|L| \leq W(\mathcal{A})$. By Lemmas 3.1 and 3.3 and (2),

$$|S| = |L| + t \leq W(\mathcal{A}) + (2^n - 2\rho(\mathcal{A})) \leq \alpha_w(Q) = \alpha(G).$$

Case 2: $L = \emptyset$. Then $\mathcal{A} = \emptyset$ and Lemma 3.1 gives $|S| = t \leq 2^n$.

Case 2a: $n \geq 3$. The set $P_{1,1} \times V(H_2) \times \dots \times V(H_n)$ is independent, so $\alpha(G) \geq \prod_{i \geq 2} |V(H_i)| \geq 3^{n-1} \geq 2^n \geq |S|$, using $3^{n-1} \geq 2^n$ for $n \geq 3$.

Case 2b: $n = 1$. Then $|S| \leq 2$ and $G = H_1$. If some part has size at least 2 then $\alpha(G) \geq 2 \geq |S|$. Otherwise G is a complete graph K_{q_1} , in which no set of size at least 2 is irredundant: a vertex v of such a set S has a neighbour in S , so it needs a private neighbour p with $N[p] \cap S = \{v\}$, but $N[p] = V(G)$. Hence $|S| \leq 1 = \alpha(G)$.

Case 2c: $n = 2$. Then $|S| \leq 4$, and $\alpha(G) \geq \max\{|V(H_1)|, |V(H_2)|\}$ by the fibres $P_{1,1} \times V(H_2)$ and $V(H_1) \times P_{2,1}$. As $q_i \geq 3$, $|V(H_i)| \geq 3$ with equality only for $H_i = K_3$; so $\alpha(G) \geq 4 \geq |S|$ unless $H_1 = H_2 = K_3$. It remains to show that $K_3 \times K_3$, whose independence number is 3, has no irredundant set of 4 social vertices. Identify $V(K_3 \times K_3)$ with the cells (r, c) of a 3×3 array; (r, c) and (r', c') are adjacent iff $r \neq r'$ and $c \neq c'$. Let S be such a set. Each $(r, c) \in S$ has an external private neighbour $(r', c') \notin S$ with $r' \neq r$, $c' \neq c$, which is non-adjacent to the other three members of S : each of them lies in row r' or in column c' ("is covered by (r', c') ").

No row contains three members of S : if $(r, a), (r, b), (r, c) \in S$ with a, b, c distinct, a private neighbour (r', c') of (r, a) has $r' \neq r$, so it must cover (r, b) and (r, c) by its column, forcing $c' = b$ and $c' = c$. Hence the numbers of members of S in the three rows are $(2, 2, 0)$ or $(2, 1, 1)$ in some order.

Pattern $(2, 2, 0)$. Say $(i, a), (i, b) \in S$ with $a \neq b$, the other two members lie in row $j \neq i$, and row k is empty. A private neighbour (r', c') of (i, a) has $r' \neq i$; to cover (i, b) it needs $c' = b$; to cover the two members in row j , which have distinct columns, it needs $r' = j$. So the private neighbour is

(j, b) , and it lies outside S . Symmetrically the private neighbour of (i, b) is $(j, a) \notin S$. Then the two members of S in row j avoid columns a and b , so both equal (j, c) for the third column c , which is absurd.

Pattern $(2, 1, 1)$. Say $(i, a), (i, b) \in S$ with $a \neq b$, and $(j, c), (k, d) \in S$ with i, j, k distinct. A private neighbour of (i, a) has row $\neq i$ and so has column b ; it covers whichever of $(j, c), (k, d)$ is not in its row only through column b , so $b \in \{c, d\}$. Likewise, from the private neighbour of (i, b) , $a \in \{c, d\}$. Hence $\{c, d\} = \{a, b\}$, and after renaming j, k we may assume $(j, a), (k, b) \in S$. Now a private neighbour (r', c') of (j, a) must cover both (i, a) and (i, b) ; a single column cannot, so $r' = i$. It must have $c' \neq a$ and cover (k, b) , which is not in row i , so $c' = b$. Thus the private neighbour is $(i, b) \in S$, contradicting the requirement that it be external (it would satisfy $\{(i, b), (j, a)\} \subseteq N[(i, b)] \cap S$).

So every irredundant set of $K_3 \times K_3$ has at most $3 = \alpha(K_3 \times K_3)$ elements, and Case 2c is complete.

In every case $|S| \leq \alpha(G)$. Hence $\text{IR}(G) \leq \alpha(G) \leq \text{IR}(G)$. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup. Its step, lemma and equation numbers refer to that writeup, not to this note (the writeup's Lemmas 1–4 are [Lemmas 2.2](#) and [3.1](#) to [3.3](#) here). The scripts it mentions are in `verification_astra/scripts/1904.02595__00/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The prompt asks for Conjecture 1.2 of Alon–Defant (arXiv:1904.02595): if $H_1, \dots, H_n$ are complete multipartite graphs and $G = \prod H_i$ (direct/tensor product), then $\alpha(G) = \text{IR}(G)$. The writeup proves exactly this statement, for arbitrary (unbalanced) complete multipartite factors and arbitrary $n \geq 1$.

Definitions were checked against the source, not taken on trust:

- The source defines S irredundant by $N[S \setminus \{v\}] \neq N[S]$ for all $v \in S$, and $\text{IR}(G)$ as the maximum size of an irredundant set. The writeup uses “for each $v \in S$ there is p with $N[p] \cap S = \{v\}$ ”. These are equivalent: $N[S \setminus \{v\}] \neq N[S]$ iff some $p \in N[v]$ lies outside $N[S \setminus \{v\}]$, i.e. $N[p] \cap S = \{v\}$. No silent weakening.
- The writeup’s reduction to “external” private neighbours is legitimate: if $p \in S$ then $p \in N[p] \cap S = \{v\}$ forces $p = v$, which is the lonely case.
- Adjacency convention for the direct product (adjacent in **every** coordinate) matches the source.

No interpretation gaming: the theorem proved is the intended conjecture, with no extra hypotheses, no restriction to balanced factors, no “as literally stated” dodge. A reasonable author of the problem would regard Conjecture 1.2 as resolved by this argument (and, as a by-product, Burcroff’s Conjecture 1.1 including the 37 cases the source paper leaves open, and the Defant–Iyer conjecture).

This is a retry target. The prompt contained an earlier gpt-5.6-sol attempt (verdict partial) but **no referee report on it**. The new writeup reuses none of the earlier attempt’s load-bearing machinery: it discards the blocker inequality (Lemma 3.2/7.1 there) and the low-dimensional classifications, and reproves the bipartite lemma from scratch. The earlier attempt’s stated obstruction (its section 8: in K_3^4 , $A = \{(0,0,0,0), (1,0,0,0)\}$ has $|A| + |B(A)| = 59 > 54 = 2\alpha$) is simply irrelevant to the new route; I checked that family explicitly — it has $\rho(A) = 2$, so Lemma 4 demands only $27 - 2 \geq 16 - 4$, which holds with room to spare. So the new writeup does not reproduce the earlier stall, and inherits nothing unchecked.

B.2 Step-by-step findings

Numbering follows the writeup. Every step was re-derived from scratch.

step	label	note
1. Setup: $\text{IR} \geq \alpha$, lonely/social split, external private neighbours	VALID	Every independent set is irredundant (take $p = v$), so only $ S \leq \alpha(G)$ is needed. Social $\Rightarrow$ external private neighbour, as argued above.

step	label	note
2. Lemma 1 (bipartite $\Rightarrow$ IR = α)	VALID	$I = (S \cap X) \cup L_Y \cup \{p_y\}$ is independent: $S \cap X$ and all p_y lie in X ; lonely vertices have no S -neighbour, so none in $S \cap X$; p_y private $\Rightarrow$ not adjacent to any lonely y' in $L_Y \subset S$. The p_y are pairwise distinct (a shared private neighbour would have two S -neighbours) and outside S , so $ I = S $.
3. Reduction: factor with 0 parts / 1 part / 2 parts	VALID	1 part $\Rightarrow$ that factor is edgeless $\Rightarrow$ G edgeless; 2 parts $\Rightarrow$ G is bipartite (colour by the side of that coordinate; adjacency forces different sides). Hence $\text{WLOG } q_i \geq 3$ for all i .
4. Quotient $Q = \prod K_{\{q_i\}}$, weights $w(x)$, identity $\alpha(G) = \alpha_w(Q)$ (2.1)	VALID	Cells are false-twin classes; two cells are completely joined iff labels differ everywhere, otherwise completely non-adjacent. Any independent set can be saturated cell-wise.
5. Tensor model $z_x = \text{tensor}(1, x_i)$, form $\beta = \text{tensor det}$, $\beta(z_x, z_y) = \prod (y_i - x_i)$ (2.2)	VALID	$\det((1, a), (1, b)) = b - a$. Non-degeneracy: the Gram matrix is $J^{\otimes n}$ with $J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, invertible. β is symmetric for even n and alternating for odd n ; in both cases $\dim W^{\perp} = 2^n - \dim W$.
6. Lemma 2: $t + 2\rho(A) \leq 2^n$ (3.1)	VALID	(a) $\beta(z_{\{y_j\}}, z_{\{x_k\}}) \neq 0$ iff $j = k$ — from privacy, and adjacency in G between cells equals adjacency of labels in Q . (b) $z_{\{y_j\}}^{\perp} \cap M(A)$: a private neighbour is non-adjacent to every lonely vertex. (c) Pairing a relation $\sum c_k z_{\{x_k\}}$ in $M(A)$ with $z_{\{y_j\}}$ gives $c_j = 0$, so $\dim(M(A) + \text{span}\{z_{\{x_k\}}\}) = \rho(A) + t$. (d) That whole space lies in $M(A)^{\perp}$: lonely vertices are pairwise non-adjacent and $\beta(z_x, z_x) = 0$, so $M(A)$ is isotropic; and a social vertex adjacent to a lonely vertex would make the latter non-lonely. (e) $\dim M(A)^{\perp} = 2^n - \rho(A)$. Hence $\rho(A) + t \leq 2^n - \rho(A)$. This is a genuine strengthening of Theorem 1.3 of the source paper ($ \text{Soc}(S) \leq 2^n$), which is the $\rho = 0$ case.
6'. (implicit) distinct social vertices occupy distinct cells	VALID (implicit)	The writeup indexes social labels $x_1, \dots, x_t$ without remarking that they are distinct. This is forced: if $x_j = x_k$ for $j \neq k$, (3.2) would assert both $\neq 0$ and $= 0$ for the same pairing. So the proof is self-consistent; a one-line remark ("false twins cannot both be social") would make it airtight-by-inspection.
7. Lemma 3(1): compression preserves independence	VALID	Not a substantive gap. If two rest-tuples differ in every coordinate $\neq i$ and both fibres are non-empty, independence forces every label of one fibre to equal every label of the other, so both fibres are the <i>same singleton</i> ; compression sends both to 1. All other pairs share a rest-coordinate.
8. Lemma 3(2): $W(C_i A) \geq W(A)$	VALID	With parts $s_{\{i,1\}} \geq \dots \geq s_{\{i,q_i\}}$, a k -element fibre is replaced by the k largest sizes. (The sorting convention, fixed in §2, is load-bearing exactly here.)
9. Lemma 3(3): $\rho(C_i A) \leq \rho(A)$	VALID	Compression is injective, so $ C_i A = A $. With $V = \mathbb{R}^2 \otimes V_{\{-i\}}$, $\pi = \text{ell} \otimes \text{id}$, $\text{ell}(a, b) = a: \pi(M) = T$ since $\text{ell}(u_{\{i,a\}}) = 1$; and $ A_x \geq 2$ gives $(u_{\{i,a\}} - u_{\{i,a'\}}) \otimes z_x = (0, a - a') \otimes z_x$ in M , so $(0, 1) \otimes E \subset M \cap \ker \pi$. Rank-nullity gives $\rho(A) \geq \dim T + \dim E$. After compression $M(C_i A) = u_{\{i,1\}} \otimes T \oplus (0, 1) \otimes E$ (the sum is direct because π kills the second summand and is injective on the first), so $\rho(C_i A) = \dim T + \dim E$.

step	label	note
10. Termination of compression; D is a down-set (4.2)	VALID	$\sum_x \sum_i x_i$ strictly drops on every effective compression. A family all of whose fibres are initial segments is a down-set (decrease one coordinate at a time).
11. $\rho(D) = D \cap \{1, 2\}^n $ (5.2)	VALID	$(1, a) = (2-a)(1, 1) + (a-1)(1, 2)$, so z_x expands over z_b with $b \in \{1, 2\}^n$, $b \leq x$; the down-set property puts those b in D . The z_b , $b \in \{1, 2\}^n$, are a basis of V .
12. $F = \{T : b(T) \in D\}$ is upward-closed, contains $[n]$, and is intersecting (5.3)–(5.4)	VALID	$T \subset U \Rightarrow b(U) \leq b(T)$; D non-empty down-set contains $(1, \dots, 1)$; disjoint T, U in F would force $[n] \setminus U$ in F and then $b(U)$, $b([n] \setminus U)$ in D differ in every coordinate, contradicting independence. The degenerate case $U = \emptyset$ is covered by the same argument.
13. $E(H)$ independent for intersecting H ; $D \subset E(F)$ (5.5)	VALID	Two members of $E(H)$ share a coordinate equal to 1. For $x \in D$ with $T = \{i : x_i = 1\}$, $b(T) \leq x$, so $b(T) \in D$ and $T \in F$.
14. Maximal intersecting families have size 2^{n-1} (5.6)	VALID	Standard; the inline proof (maximal $\Rightarrow$ upward-closed $\Rightarrow$ exactly one of each complementary pair) is correct.
15. $m(T) \geq 2$ for $T \in F^* \setminus F$ (5.7)–(5.8)	VALID	$[n] \in F$, so such a T misses some i , and $\sum_{a \geq 2} s_{\{i, a\}} \geq q_i - 1 \geq 2$ since $q_i \geq 3$. This is the only place $q_i \geq 3$ is used in Lemma 4.
16. Lemma 4: $\alpha_w(Q) - W(A) \geq 2^n - 2\rho(A)$ (5.1), (5.9)	VALID	$W(E(F^*)) - W(E(F)) = \sum_{T \in F^* \setminus F} m(T) \geq 2(2^{n-1} - F) = 2^n - 2\rho(D)$, and $\alpha_w \geq W(E(F^*)) - W(E(F)) \geq W(D) \geq W(A)$, $\rho(D) \leq \rho(A)$. The chain $\alpha_w - W(A) \geq W(E(F^*)) - W(E(F))$ is correct because $\alpha_w - W(E(F^*)) \geq 0 \geq W(A) - W(E(F))$.
17. §6: irredundant sets with a lonely vertex (6.1)	VALID	A (lonely support) is non-empty and independent in Q (lonely vertices are pairwise non-adjacent), $ L \leq W(A)$, and Lemmas 2+4 give $ S = L + t \leq W(A) + 2^n - 2\rho(A) \leq \alpha_w = \alpha(G)$. The two inequalities fit exactly, as claimed.
18. §7.1: $n \geq 3$, lonely-free $\Rightarrow S \leq 2^n - 3^{n-1} \leq \alpha(G)$	VALID	$\alpha(G) \geq P_{\{1, 1\}} * \prod_{i \geq 2} V(H_i) \geq 3^{n-1}$, and $3^{n-1} \geq 2^n$ for $n \geq 3$.
19. §7.2: $n = 1$	VALID	$ S \leq 2$; if some part has size ≥ 2 then $\alpha \geq 2$; otherwise G is complete and no irredundant set has ≥ 2 vertices ($N[p]$ is everything).
20. §7.3: $n = 2$ reduces to $K_3 \times K_3$, and $K_3 \times K_3$ has no 4-element all-social irredundant set	VALID	$\alpha \geq \max(V(H_1) , V(H_2)) \geq 4$ unless both factors are K_3 . The $K_3 \times K_3$ case analysis is correct: no row holds 3 selected cells (a private neighbour of one would need two distinct columns); pattern (2,2,0) forces row j to avoid both columns a, b ; pattern (2,1,1) forces $\{c, d\} = \{a, b\}$ and then the private neighbour of (j, a) is the selected cell (i, b) . I confirmed $IR(K_3 \times K_3) = 3 = \alpha$ by exhaustive search.
21. §8 conclusion	VALID	Case coverage is complete: empty/edgeless, a factor with ≤ 2 parts (bipartite), $q_i \geq 3$ with a lonely vertex, $q_i \geq 3$ lonely-free split by $n = 1, 2, \geq 3$.

No step is labelled GAP or ERROR.

B.3 Reference check

- **arXiv:1904.02595 (Alon–Defant), fetched via ar5iv.** Confirmed verbatim: irredundant is $N[S \setminus \{v\}] \neq N[S]$; $IR(G)$ is the maximum size of an irredundant set; **Conjecture 1.1** (Burcroff, balanced factors $K[u_i, t_i]$) and **Conjecture 1.2** (“If $H_1, \dots, H_n$ are complete multipartite graphs and $G = \prod H_i$, then $\alpha(G) = IR(G)$ ”) are as quoted in the prompt. The target statement is therefore correctly transcribed.
- **Theorem 1.3 of the source:** “there exist $Lon(S)$, $Soc(S) \subset S$ such that $Lon(S)$ is independent and $|Soc(S)| \leq 2^n$. In particular $IR(G) \leq \alpha(G) + 2^n$.” It is proved by the polynomial method: $f_v(x_1, \dots, x_n) = \prod_i (x_i - \mathbb{1}_{v(i)})$ inside

the 2^n -dimensional space of multilinear polynomials. This is the *dual* formulation of the writeup’s tensor model, and the writeup’s Lemma 2 is a strict refinement of it (the $\rho(A) = 0$ case). The writeup does not cite the source for this; it proves it afresh, which is legitimate, but see Caveats on attribution.

- **The 37 exceptional cases** of the source (fetched): $K_{3^4}, K_{2,3} \times K_{3^3}, K_4 \times K_{3^3}, K_5 \times K_4 \times K_{3^2}, \dots, K_{3^5}, K_{3^6}, K_4 \times K_{3^5}$. The new argument covers all of them; I ran the load-bearing inequality on K_{3^4} and K_{3^5} specifically (below).
- **External results invoked**: essentially none. The only classical fact is “a maximal intersecting family on $[n]$ has size 2^{n-1} ”, which the writeup proves inline (correctly). Everything else is self-contained linear algebra. So there is no misquoted-citation failure mode available here.
- **Priority search** (web): no paper resolving Conjecture 1.2 was found (searches for follow-ups on Burcroff’s conjecture / upper irredundance in direct products, 2019–2026 returned only the source paper, its journal version in *Discrete Math.* 343 (2020), and unrelated domination work). The catalog’s May-2026 review reached the same conclusion. The result therefore appears to be new, not ALREADY_KNOWN.

B.4 Computational check

Scripts in `verification_astra/scripts/1904.02595__00/` (`core.py`, `check_ir_alpha.py`, `check_lemmas.py`, `check_lemma4.py`, `check_lemma4_n34.py`, `stress_lemma4.py`, `check_lemma2_k33.py`, `stress_lemma2.py`, `stress_lemma2_n4.py`, `check_misc.py`, `check_k33x3.py`). IR is computed by exhaustive DFS over irredundant sets (irredundance is hereditary, so the search is complete); alpha by max-weight clique on the complement of the quotient.

1. **IR = alpha by brute force.** 76 distinct products of complete multipartite graphs with $|V| \leq 18$ (1, 2 and 3 factors, balanced and unbalanced, e.g. $K(2,2,1) \times K(2,1), K(3,1,1) \times K(1,1,1), K(2,1) \times K(1,1) \times K(1,1,1)$): **IR = alpha in every case, 0 mismatches.** Plus three larger exhaustive runs: $K_3 \times K_3 \times K_3$ (27 vertices, 207,244 irredundant sets, $\alpha = \text{IR} = 9$), $K(2,1,1) \times K_3 \times K_2$ (24 vertices, 140,016 sets, $\alpha = \text{IR} = 12$), $K(2,2) \times K_3 \times K_3$ (36 vertices, 4,223,278 sets, $\alpha = \text{IR} = 18$), and $K(2,1,1) \times K_3 \times K_3$ (36 vertices, 2,020,728 sets, unbalanced, $\alpha = \text{IR} = 18$). In particular $\text{IR}(K_3 \times K_3) = 3$, confirming §7.3.
2. **Lemma 2 over all irredundant sets.** For 9 small products, and separately for *all* 207,244 irredundant sets of K_{3^3} , the quantity $2^n - t - 2\rho(A)$ was computed exactly (rank over the rationals via sympy / numerically via numpy): **minimum 0, no violation.** The bound is tight (e.g. a full “row” of $K_3 \times K_3$: $t = 0, \rho = 2, 2^n = 4$).
3. **Lemma 2 adversarially, at the quotient level.** I searched directly over combinatorial configurations $(A, x_1 \dots x_t, y_1 \dots y_t)$ satisfying exactly the privacy constraints used in the proof, maximising t by maximum clique on admissible (x, y) pairs: all A in K_{3^2} (34 families), all $|A| \leq 2$ in K_{3^3} (271) plus random larger A , and 300 random A in K_{3^4} with greedy t . **Minimum slack 0, no violation.** (Tightness note: in K_{3^2} the relaxed quotient configuration attains $t = 4 = 2^n$ with A empty — which is precisely why §7.3 needs a separate argument for $K_3 \times K_3$; there the “private neighbour” would have to be a selected vertex. The writeup handles exactly this.)
4. **Lemma 3 (compression).** 300 random families in q_i in $\{3, 4\}$, $n \leq 4$: full compression always terminated in a down-set, preserved cardinality, did not decrease weight, and did not increase rank — **0 problems**; same checks inside the Lemma-4 sweeps, **0 violations.**
5. **Eq. (5.2).** $\rho(D) = |D \cap \{1, 2\}^n|$ on 400 random down-sets ($n \leq 4, q \leq 5$): **0 mismatches.**
6. **Lemma 4 exhaustively.** All non-empty independent families with unit and with random part sizes: $n = 1$ ($q = 3, 4, 5$), $n = 2$ ($q = (3, 3), (3, 4), (4, 4)$: 33, 61, 104 families each, 4 weightings each); $n = 3$ all families of size ≤ 3 (1215 for K_{3^3} , 2478 for $q = (3, 4, 3)$) plus random maximal ones; $n = 4$ all families of size ≤ 2 (2673) plus random maximal ones. **0 violations; minimum slack 0**, attained exactly at maximum independent families, where $\rho = 2^{n-1}$ — i.e. the inequality is tight precisely where the proof predicts.
7. **The source paper’s open cases.** For K_{3^4} and K_{3^5} (on the list of 37), Lemma 4 with unit weights is equivalent to $\max(|A| - 2\rho(A)) \leq \alpha - 2^n$, i.e. ≤ 11 and ≤ 49 . Simulated annealing plus random maximal families plus the canonical families found maxima of exactly **11** and **49** (attained by a coordinate fibre, $|A| = 27, \rho = 8$ and $|A| = 81, \rho = 16$) — tight, no violation. A seeded local search for a large irredundant set in K_{3^4} (81 vertices) found nothing beyond $|S| = 27 = \alpha$.

No computational check failed; nothing in the writeup rests on an unverified finite object.

B.5 Caveats

1. **Attribution/priority, not correctness.** Lemma 2 with $\rho(A) = 0$ is Theorem 1.3 of the source paper (proved there by the polynomial method). The writeup states “the proof below is self-contained” and “literature priority is unchecked”; for publication it must cite Alon–Defant Theorem 1.3 and present Lemma 2 as a refinement of it. The genuinely new ingredients are the $2\rho(A)$ term and Lemma 4.
2. **ρ depends on the labelling.** $\rho(A)$ is defined through the specific embedding $a \mapsto (1, a)$ with parts indexed in *decreasing size order*. Lemmas 2 and 4 use the same fixed labelling, so the combination in §6 is legitimate — but the writeup never remarks that ρ is not a labelling-invariant of A , and the sorted-order convention (fixed in §2) is silently re-used in Lemma 3(2). Both are expository, not mathematical, gaps.
3. **Implicit step in Lemma 2:** distinct social vertices lie in distinct cells (false twins cannot both be social). This is forced by the displayed relations (3.2), so nothing breaks, but the writeup should say it.
4. $q_i \geq 3$ **scope.** Lemma 4 and §7.1/§7.3 assume every factor has at least three parts. This is justified only by the earlier reduction (a factor with ≤ 2 parts makes the whole product edgeless or bipartite). Note the reduction does not delete the small factor, it disposes of the whole product — which is correct, but means Lemma 4 is genuinely unavailable for products with a bipartite factor; those are covered by Lemma 1 alone.
5. **Edge cases.** $n = 0$ (empty product) and factors with no vertices are dismissed in one line; both are trivially fine ($\alpha = \text{IR} = 0$ or 1). $n = 1$ and $n = 2$ need the separate §7.2/§7.3 arguments, and $K_3 \times K_3$ genuinely is the unique hard lonely-free case — the writeup finds it.
6. **Strength of the claim.** The theorem, if correct, resolves Alon–Defant Conjecture 1.2, hence also Burcroff’s Conjecture 1.1 *including the 37 cases the source paper could not settle*, and the Defant–Iyer conjecture, by an argument of three pages that needs no isoperimetry or stability. I could not find an error and every load-bearing inequality survives exhaustive computational attack, including in

the exceptional cases themselves; but a result that supersedes a published Alon paper's main theorem this cheaply should get independent human verification before submission.

7. The verdict block's one-liner ("a tensor-rank bound retaining the lonely vertices, combined with rank-nonincreasing coordinate compression") accurately describes what is proved; there is no gap between the announced claim and the theorem.

B.6 Referee summary

I set out to break this writeup and could not. The argument is short and entirely self-contained, and it survives a line-by-line re-derivation: the quotient reduction, the tensor/determinant-form model, the refinement $t + 2 \rho(A) \leq 2^n$ of Alon–Defant's Theorem 1.3 (whose proof correctly uses that $M(A)$ is isotropic and that private neighbours annihilate it), the three compression claims (independence, weight, rank — the rank part via $\rho \geq \dim T + \dim E$ and the exact identity after compression), the down-set/intersecting-family identification $\rho(D) = |D \cap \{1, 2\}^n| = |F|$, and the weighted rank-defect inequality $\alpha_w - W(A) \geq 2^n - 2 \rho(A)$ obtained by extending F to a maximal intersecting family and charging at least 2 to each new pattern class. The two inequalities combine exactly, and the residual lonely-free cases ($n = 1, K_3 \times K_3, n \geq 3$ via $3^{n-1} \geq 2^n$) are handled correctly, the $K_3 \times K_3$ case analysis being the only fiddly point — and brute force confirms $IR(K_3 \times K_3) = 3$. Computationally, $IR = \alpha$ held in 80 exhaustively searched products (up to 36 vertices and 4.2M irredundant sets), Lemma 2 held over all 207,244 irredundant sets of K_3^3 and over adversarially constructed quotient configurations, and Lemma 4 held in every exhaustive and randomized sweep, tight exactly where the proof predicts — including in K_3^4 and K_3^5 , two of the 37 cases the source paper leaves open. The only defects I found are expository (an implicit "distinct social vertices occupy distinct cells", no remark that ρ is labelling-dependent) and one of scholarship: Lemma 2 at $\rho = 0$ is the published Theorem 1.3 of arXiv:1904.02595 and must be cited. Verdict: CONFIRMED.

References

- [1] N. Alon and C. Defant, Isoperimetry, stability, and irredundance in direct products, *Discrete Math.* 343 (2020), no. 7, 111902; [arXiv:1904.02595](#).
- [2] A. Burcroff, Domination parameters of the unitary Cayley graph of $\mathbb{Z}/n\mathbb{Z}$, *Discuss. Math. Graph Theory* 43 (2023), 95–114; [arXiv:1809.04769](#).
- [3] C. Defant and S. Iyer, Domination and upper domination of direct product graphs, *Discrete Math.* 341 (2018), 2742–2752; [arXiv:1708.01305](#).